Group 6

Gender Differences in Weight and Height

Research Question

1. Are male students heavier than female students?
2. Are male students taller than female students?
3. Is the difference statistically significant?

Methods

Independence

Welch's Two-Sample t-test

Normality

Homoscedasticity

Welch's t-test was validated and selected due to large sample sizes and independent random sampling

Summary

Male adolescents are significantly heavier and taller than females.

Data

Dataset：YRBS 2007

Variables

- Grouping1:Male(n=7510)
- Group 2: Female (n=6166)

Response Variables

- Weight (kg)
- Height (cm)

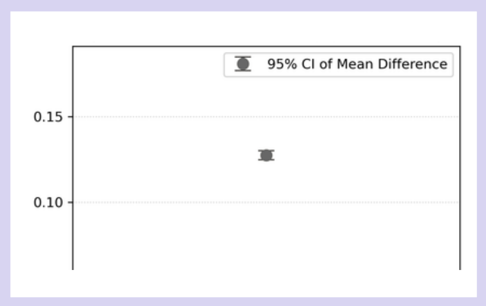
Sample Size

- Total ：14,041
- Weight available: Male = 5510 Female = 6166
- Height available: Male = 6490 Female = 6572

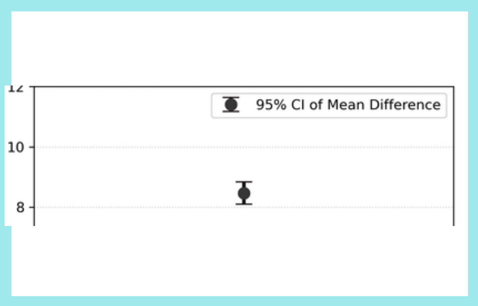
EDA Highlights

- Males are 12.75 cm taller than females on average.
- Males weigh 8.47 kg more than females on average.
- Males weigh visibly more than females overall.

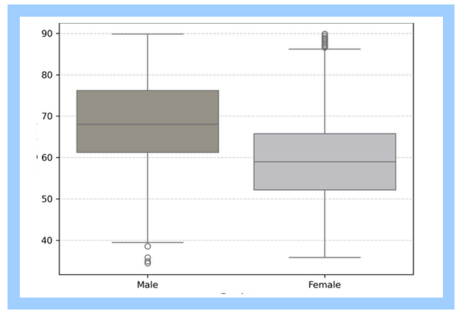
95% CI for Height Difference



95% CI for Weight Difference



Weight Distribution by Gender



Results

$$H_0 : \mu_{weight,M} = \mu_{weight,F}$$
$$H_1 : \mu_{weight,M} \neq \mu_{weight,F}$$

$$H_0 : \mu_{height,M} = \mu_{height,F}$$
$$H_1 : \mu_{height,M} \neq \mu_{height,F}$$

Weight (kg)

Male: $SD = 10.25$, $\bar{X} = 60.23$

Female: $SD = 10.23$, $\bar{X} = 68.70$

$t = 44.62, p < 0.0001$ (Reject H_0)

Mean Difference = 8.47 kg

$95\%CI : [8.10, 8.84] kg$

Males are significantly heavier.

Height (cm)

Male: $SD = 8$, $\bar{X} = 176$

Female: $SD = 7$, $\bar{X} = 163$

$t = 92.31, p < 0.0001$ (Reject H_0)

Mean Difference = 12.75cm

$95\%CI : [12.47, 13.02] cm$

Males are significantly taller.

ALL TESTS

✓ Reject H_0

✓ $p < 0.0001$

✓ Highly significant gender differences

Effect Size

- Weight : d = 0.83 Large
- Height : d = 1.67 Very Large

Limitation

- Observational data
- Correlation only
- Not causal